Likhith Kumar

Unity Game Developer · Multiplayer Systems · Gameplay Programming · Physics & AI

+91 93538 36006 | likhithkulal0@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Results-driven Unity Game Developer with hands-on experience building **real-time multiplayer systems**, **physics-based gameplay**, and **scalable game architectures** using C# and Photon Fusion. Proven track record of delivering high-performance 2D/3D game systems at a professional game studio, with expertise in performance optimization, AI behaviors, and clean modular code design. Passionate about crafting responsive, engaging player experiences with a strong foundation in software engineering principles and design patterns.

Technical Skills

Languages	C#, C++, Object-Oriented Programming, SOLID Principles
Game Engine	Unity 2D/3D, Physics, Cinemachine, Animations, Timeline, Addressables, UI Toolkit
Multiplayer	Photon Fusion 2, Client-Server Architecture, State Synchronization, Matchmaking & Real Time Backend system
Architecture	Design Patterns, ScriptableObjects, Event-Driven Systems, Unity DOTS / ECS (Basics)
AI & Tools	NavMesh Agent, Behavior Trees, Finite State Machines, Git & GitHub, Unity Profiler, Rider IDE
Optimization	Object Pooling, Draw Call Batching, Memory Profiling, LOD Systems, Frame Rate Stabilization

Professional Experience

Unity Game Developer · *Juego Studios Pvt. Ltd.* **Feb 2025 – Present**

Bangalore, India · Game Development Studio

- Architected and shipped scalable **2D and 3D gameplay systems** in Unity/C# across multiple live projects, improving code modularity and reducing iteration time for the team.
- Integrated **Photon Fusion 2** to implement real-time multiplayer features including matchmaking, networked object lifecycle, client-side prediction, and state synchronization for up to 10 concurrent players.
- Engineered **physics-based gameplay mechanics** using Rigidbody, Joints, and custom force models to deliver stable, responsive player interactions with minimal jitter.
- Designed **event-driven, decoupled architectures** using ScriptableObjects and the Observer pattern, enabling teams to extend gameplay systems with zero coupling between modules.
- Achieved measurable frame rate improvements through systematic **profiling and optimization**: implemented object pooling, draw call batching, and memory management strategies.
- Developed **NavMesh-based AI agents** with state-machine behavior trees, resulting in dynamic, context-aware NPC interactions that enhanced gameplay depth.
- Built **responsive UI/HUD systems** using Unity UI Toolkit and Canvas, supporting multiple aspect ratios and screen resolutions across target platforms.
- Collaborated in an Agile environment with designers, artists, and producers to convert gameplay requirements into production-ready, documented technical solutions.

Key Projects

CarLeague — Real-Time Multiplayer Vehicle Soccer Game *Unity, Photon Fusion 2, C#*

- Engineered a **Rocket League-inspired multiplayer game** with physics-driven vehicle movement, boost mechanics, and accurate ball physics using custom Rigidbody constraints.
- Implemented **networked game state management** (goal events, match timer, score sync) across all clients with sub-50ms latency using Photon Fusion's server-authoritative model.
- Built a **matchmaking and lobby system** supporting room creation, player ready-up flow, and graceful disconnect/reconnect handling.
- Designed **arcade-tuned vehicle physics** with camera shake, collision feedback, and responsive controls that maintained a consistent feel regardless of network conditions.

Advanced Character Controller *Unity, C#, Physics*

- Built a fully **custom physics-based character controller** from scratch — bypassing Unity's CharacterController — for precise control over movement feel, step handling, and slope detection.
- Implemented a **hierarchical state machine** (idle, run, sprint, jump, air, fall, wall-jump) with clean transition logic, enabling rapid addition of new states without regression.
- Integrated **camera-relative directional input** and configurable movement parameters (acceleration curves, coyote time, jump buffering) as a reusable, project-agnostic system.

Royal Run — Mobile Endless Runner *Unity, C#, Addressables*

- Designed a **procedural level generation system** using weighted tile selection and difficulty scaling to create infinitely varied, replayable gameplay.
- Implemented a **high-performance object pooling system** that reduced garbage collection spikes by over 60%, maintaining stable 60 FPS on mid-range devices.
- Built a complete **meta-game loop**: power-up system, combo multiplier, high-score tracking, and animated UI with TextMeshPro and Unity Canvas.

Education

B.E. Computer Science & Engineering — *AJ Institute of Engineering and Technology, Mangalore* May 2025
CGPA: **9.0 / 10.0**

Certifications & Professional Development

- **Unity 3D Game Development** — Core gameplay programming and engine fundamentals
- **Unity Artificial Intelligence** — NavMesh, pathfinding, and NPC behavior systems
- **Design Principles in Game Development** — Game feel, systems design, and balancing
- **Git & GitHub** — Version control best practices for team-based development
- **AI for Developers** — Workflow acceleration with Cursor AI and GitHub Copilot